

LIFEWS AgriShine(TM) GrowMeal(TM)

JSS 1-3 ASSESSMENT & QUIZ SYSTEM

TEACHER MARKING & MODERATION GUIDE

Purpose: assess knowledge, practical competence, investigation/data skills, and mission-based problem solving without reducing garden learning to recall alone.

Domain	Weight	Primary evidence
Knowledge	15%	Oral/written response, concept explanation
Practical competence	25%	Demonstration, technique, safety, record quality
Investigation/Data	30%	Measurement, table/graph, calculation, interpretation
Mission problem solving	30%	Diagnosis, design, recommendation, evidence-based decision

Performance rubric for open/practical items

Score	Descriptor
3 - Proficient	Method is appropriate, evidence is accurate/traceable, and explanation or decision is supported.
2 - Developing	Method is mostly appropriate with minor errors or incomplete reasoning.
1 - Beginning	Major support is needed; evidence or reasoning is incomplete.
0 - Not demonstrated	No valid evidence or the task is not completed.

Moderation rule: award credit for scientifically or practically valid alternatives when they satisfy the task. Do not award marks for unsupported claims. For Nursery, use oral responses and observed performance where writing is not developmentally appropriate.

ANSWER & MARKING GUIDE 01 - Production Garden

AQ-301 - Knowledge

State a testable hypothesis related to establishment, growth, yield and input use.

Expected answer / marking basis: Hypothesis names a predictor/treatment and measurable response.

AQ-302 - Knowledge

Differentiate the response variable from one controlled variable in a study of the production plot.

Expected answer / marking basis: Correct variable classification.

AQ-303 - Knowledge

Why are replication and traceable records important?

Expected answer / marking basis: They characterize variation and allow checking/reproducibility.

AQ-304 - Practical

Set up or demonstrate a teacher-approved measurement of production metric, including unit and sample ID.

Expected answer / marking basis: 0-3 technical performance rubric.

AQ-305 - Practical

Perform one data-quality check appropriate to the measurement.

Expected answer / marking basis: 0-3 QC rubric.

AQ-306 - Investigation/Data

Analyze replicate data using an appropriate summary and quantify variation.

Expected answer / marking basis: Correct summary + variation measure.

AQ-307 - Investigation/Data

Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

Expected answer / marking basis: Correct comparison and effect statement.

AQ-308 - Mission

Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

Expected answer / marking basis: Plausible diagnosis + confirmatory evidence.

AQ-309 - Mission

Write a qualified conclusion that includes uncertainty or a limitation.

Expected answer / marking basis: Conclusion matches data and qualifies certainty.

AQ-310 - Mission

Recommend a management decision and define the evidence threshold that would trigger it.

Expected answer / marking basis: Decision + measurable trigger.

ANSWER & MARKING GUIDE 02 - Raised Systems Garden

AQ-311 - Knowledge

State a testable hypothesis related to root-zone condition, volume, infiltration and productivity.

Expected answer / marking basis: Hypothesis names a predictor/treatment and measurable response.

AQ-312 - Knowledge

Differentiate the response variable from one controlled variable in a study of the raised system.

Expected answer / marking basis: Correct variable classification.

AQ-313 - Knowledge

Why are replication and traceable records important?

Expected answer / marking basis: They characterize variation and allow checking/reproducibility.

AQ-314 - Practical

Set up or demonstrate a teacher-approved measurement of system metric, including unit and sample ID.

Expected answer / marking basis: 0-3 technical performance rubric.

AQ-315 - Practical

Perform one data-quality check appropriate to the measurement.

Expected answer / marking basis: 0-3 QC rubric.

AQ-316 - Investigation/Data

Analyze replicate data using an appropriate summary and quantify variation.

Expected answer / marking basis: Correct summary + variation measure.

AQ-317 - Investigation/Data

Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

Expected answer / marking basis: Correct comparison and effect statement.

AQ-318 - Mission

Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

Expected answer / marking basis: Plausible diagnosis + confirmatory evidence.

AQ-319 - Mission

Write a qualified conclusion that includes uncertainty or a limitation.

Expected answer / marking basis: Conclusion matches data and qualifies certainty.

AQ-320 - Mission

Recommend a management decision and define the evidence threshold that would trigger it.

Expected answer / marking basis: Decision + measurable trigger.

ANSWER & MARKING GUIDE 03 - Grid Experiment Garden

AQ-321 - Knowledge

State a testable hypothesis related to treatment-control comparisons and fair testing.

Expected answer / marking basis: Hypothesis names a predictor/treatment and measurable response.

AQ-322 - Knowledge

Differentiate the response variable from one controlled variable in a study of the grid cell.

Expected answer / marking basis: Correct variable classification.

AQ-323 - Knowledge

Why are replication and traceable records important?

Expected answer / marking basis: They characterize variation and allow checking/reproducibility.

AQ-324 - Practical

Set up or demonstrate a teacher-approved measurement of treatment response, including unit and sample ID.

Expected answer / marking basis: 0-3 technical performance rubric.

AQ-325 - Practical

Perform one data-quality check appropriate to the measurement.

Expected answer / marking basis: 0-3 QC rubric.

AQ-326 - Investigation/Data

Analyze replicate data using an appropriate summary and quantify variation.

Expected answer / marking basis: Correct summary + variation measure.

AQ-327 - Investigation/Data

Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

Expected answer / marking basis: Correct comparison and effect statement.

AQ-328 - Mission

Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

Expected answer / marking basis: Plausible diagnosis + confirmatory evidence.

AQ-329 - Mission

Write a qualified conclusion that includes uncertainty or a limitation.

Expected answer / marking basis: Conclusion matches data and qualifies certainty.

AQ-330 - Mission

Recommend a management decision and define the evidence threshold that would trigger it.

Expected answer / marking basis: Decision + measurable trigger.

ANSWER & MARKING GUIDE 04 - Crop Rotation Garden

AQ-331 - Knowledge

State a testable hypothesis related to crop-family sequence and soil cover.

Expected answer / marking basis: Hypothesis names a predictor/treatment and measurable response.

AQ-332 - Knowledge

Differentiate the response variable from one controlled variable in a study of the rotation record.

Expected answer / marking basis: Correct variable classification.

AQ-333 - Knowledge

Why are replication and traceable records important?

Expected answer / marking basis: They characterize variation and allow checking/reproducibility.

AQ-334 - Practical

Set up or demonstrate a teacher-approved measurement of rotation interval, including unit and sample ID.

Expected answer / marking basis: 0-3 technical performance rubric.

AQ-335 - Practical

Perform one data-quality check appropriate to the measurement.

Expected answer / marking basis: 0-3 QC rubric.

AQ-336 - Investigation/Data

Analyze replicate data using an appropriate summary and quantify variation.

Expected answer / marking basis: Correct summary + variation measure.

AQ-337 - Investigation/Data

Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

Expected answer / marking basis: Correct comparison and effect statement.

AQ-338 - Mission

Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

Expected answer / marking basis: Plausible diagnosis + confirmatory evidence.

AQ-339 - Mission

Write a qualified conclusion that includes uncertainty or a limitation.

Expected answer / marking basis: Conclusion matches data and qualifies certainty.

AQ-340 - Mission

Recommend a management decision and define the evidence threshold that would trigger it.

Expected answer / marking basis: Decision + measurable trigger.

ANSWER & MARKING GUIDE 05 - Intercropping Garden

AQ-341 - Knowledge

State a testable hypothesis related to crop density, canopy, component yield and LER.

Expected answer / marking basis: Hypothesis names a predictor/treatment and measurable response.

AQ-342 - Knowledge

Differentiate the response variable from one controlled variable in a study of the intercrop plot.

Expected answer / marking basis: Correct variable classification.

AQ-343 - Knowledge

Why are replication and traceable records important?

Expected answer / marking basis: They characterize variation and allow checking/reproducibility.

AQ-344 - Practical

Set up or demonstrate a teacher-approved measurement of intercrop metric, including unit and sample ID.

Expected answer / marking basis: 0-3 technical performance rubric.

AQ-345 - Practical

Perform one data-quality check appropriate to the measurement.

Expected answer / marking basis: 0-3 QC rubric.

AQ-346 - Investigation/Data

Analyze replicate data using an appropriate summary and quantify variation.

Expected answer / marking basis: Correct summary + variation measure.

AQ-347 - Investigation/Data

Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

Expected answer / marking basis: Correct comparison and effect statement.

AQ-348 - Mission

Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

Expected answer / marking basis: Plausible diagnosis + confirmatory evidence.

AQ-349 - Mission

Write a qualified conclusion that includes uncertainty or a limitation.

Expected answer / marking basis: Conclusion matches data and qualifies certainty.

AQ-350 - Mission

Recommend a management decision and define the evidence threshold that would trigger it.

Expected answer / marking basis: Decision + measurable trigger.

ANSWER & MARKING GUIDE 06 - Water-Smart Garden

AQ-351 - Knowledge

State a testable hypothesis related to infiltration, irrigation and water productivity.

Expected answer / marking basis: Hypothesis names a predictor/treatment and measurable response.

AQ-352 - Knowledge

Differentiate the response variable from one controlled variable in a study of the water-smart bed.

Expected answer / marking basis: Correct variable classification.

AQ-353 - Knowledge

Why are replication and traceable records important?

Expected answer / marking basis: They characterize variation and allow checking/reproducibility.

AQ-354 - Practical

Set up or demonstrate a teacher-approved measurement of water metric, including unit and sample ID.

Expected answer / marking basis: 0-3 technical performance rubric.

AQ-355 - Practical

Perform one data-quality check appropriate to the measurement.

Expected answer / marking basis: 0-3 QC rubric.

AQ-356 - Investigation/Data

Analyze replicate data using an appropriate summary and quantify variation.

Expected answer / marking basis: Correct summary + variation measure.

AQ-357 - Investigation/Data

Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

Expected answer / marking basis: Correct comparison and effect statement.

AQ-358 - Mission

Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

Expected answer / marking basis: Plausible diagnosis + confirmatory evidence.

AQ-359 - Mission

Write a qualified conclusion that includes uncertainty or a limitation.

Expected answer / marking basis: Conclusion matches data and qualifies certainty.

AQ-360 - Mission

Recommend a management decision and define the evidence threshold that would trigger it.

Expected answer / marking basis: Decision + measurable trigger.

ANSWER & MARKING GUIDE 07 - Compost & Soil Garden

AQ-361 - Knowledge

State a testable hypothesis related to decomposition and soil response.

Expected answer / marking basis: Hypothesis names a predictor/treatment and measurable response.

AQ-362 - Knowledge

Differentiate the response variable from one controlled variable in a study of the compost/soil sample.

Expected answer / marking basis: Correct variable classification.

AQ-363 - Knowledge

Why are replication and traceable records important?

Expected answer / marking basis: They characterize variation and allow checking/reproducibility.

AQ-364 - Practical

Set up or demonstrate a teacher-approved measurement of soil/compost metric, including unit and sample ID.

Expected answer / marking basis: 0-3 technical performance rubric.

AQ-365 - Practical

Perform one data-quality check appropriate to the measurement.

Expected answer / marking basis: 0-3 QC rubric.

AQ-366 - Investigation/Data

Analyze replicate data using an appropriate summary and quantify variation.

Expected answer / marking basis: Correct summary + variation measure.

AQ-367 - Investigation/Data

Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

Expected answer / marking basis: Correct comparison and effect statement.

AQ-368 - Mission

Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

Expected answer / marking basis: Plausible diagnosis + confirmatory evidence.

AQ-369 - Mission

Write a qualified conclusion that includes uncertainty or a limitation.

Expected answer / marking basis: Conclusion matches data and qualifies certainty.

AQ-370 - Mission

Recommend a management decision and define the evidence threshold that would trigger it.

Expected answer / marking basis: Decision + measurable trigger.

ANSWER & MARKING GUIDE 08 - Soilless Learning Garden

AQ-371 - Knowledge

State a testable hypothesis related to recirculating flow, pH, EC and root-zone condition.

Expected answer / marking basis: Hypothesis names a predictor/treatment and measurable response.

AQ-372 - Knowledge

Differentiate the response variable from one controlled variable in a study of the soilless unit.

Expected answer / marking basis: Correct variable classification.

AQ-373 - Knowledge

Why are replication and traceable records important?

Expected answer / marking basis: They characterize variation and allow checking/reproducibility.

AQ-374 - Practical

Set up or demonstrate a teacher-approved measurement of solution metric, including unit and sample ID.

Expected answer / marking basis: 0-3 technical performance rubric.

AQ-375 - Practical

Perform one data-quality check appropriate to the measurement.

Expected answer / marking basis: 0-3 QC rubric.

AQ-376 - Investigation/Data

Analyze replicate data using an appropriate summary and quantify variation.

Expected answer / marking basis: Correct summary + variation measure.

AQ-377 - Investigation/Data

Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

Expected answer / marking basis: Correct comparison and effect statement.

AQ-378 - Mission

Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

Expected answer / marking basis: Plausible diagnosis + confirmatory evidence.

AQ-379 - Mission

Write a qualified conclusion that includes uncertainty or a limitation.

Expected answer / marking basis: Conclusion matches data and qualifies certainty.

AQ-380 - Mission

Recommend a management decision and define the evidence threshold that would trigger it.

Expected answer / marking basis: Decision + measurable trigger.

ANSWER & MARKING GUIDE 09 - Sensor-Ready Garden

AQ-381 - Knowledge

State a testable hypothesis related to sensor repeatability, error, placement and drift.

Expected answer / marking basis: Hypothesis names a predictor/treatment and measurable response.

AQ-382 - Knowledge

Differentiate the response variable from one controlled variable in a study of the sensor station.

Expected answer / marking basis: Correct variable classification.

AQ-383 - Knowledge

Why are replication and traceable records important?

Expected answer / marking basis: They characterize variation and allow checking/reproducibility.

AQ-384 - Practical

Set up or demonstrate a teacher-approved measurement of sensor metric, including unit and sample ID.

Expected answer / marking basis: 0-3 technical performance rubric.

AQ-385 - Practical

Perform one data-quality check appropriate to the measurement.

Expected answer / marking basis: 0-3 QC rubric.

AQ-386 - Investigation/Data

Analyze replicate data using an appropriate summary and quantify variation.

Expected answer / marking basis: Correct summary + variation measure.

AQ-387 - Investigation/Data

Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

Expected answer / marking basis: Correct comparison and effect statement.

AQ-388 - Mission

Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

Expected answer / marking basis: Plausible diagnosis + confirmatory evidence.

AQ-389 - Mission

Write a qualified conclusion that includes uncertainty or a limitation.

Expected answer / marking basis: Conclusion matches data and qualifies certainty.

AQ-390 - Mission

Recommend a management decision and define the evidence threshold that would trigger it.

Expected answer / marking basis: Decision + measurable trigger.

ANSWER & MARKING GUIDE 10 - Applied Systems Garden

AQ-391 - Knowledge

State a testable hypothesis related to stocks, flows, feedbacks, trade-offs and scenarios.

Expected answer / marking basis: Hypothesis names a predictor/treatment and measurable response.

AQ-392 - Knowledge

Differentiate the response variable from one controlled variable in a study of the system map.

Expected answer / marking basis: Correct variable classification.

AQ-393 - Knowledge

Why are replication and traceable records important?

Expected answer / marking basis: They characterize variation and allow checking/reproducibility.

AQ-394 - Practical

Set up or demonstrate a teacher-approved measurement of system indicator, including unit and sample ID.

Expected answer / marking basis: 0-3 technical performance rubric.

AQ-395 - Practical

Perform one data-quality check appropriate to the measurement.

Expected answer / marking basis: 0-3 QC rubric.

AQ-396 - Investigation/Data

Analyze replicate data using an appropriate summary and quantify variation.

Expected answer / marking basis: Correct summary + variation measure.

AQ-397 - Investigation/Data

Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

Expected answer / marking basis: Correct comparison and effect statement.

AQ-398 - Mission

Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

Expected answer / marking basis: Plausible diagnosis + confirmatory evidence.

AQ-399 - Mission

Write a qualified conclusion that includes uncertainty or a limitation.

Expected answer / marking basis: Conclusion matches data and qualifies certainty.

AQ-400 - Mission

Recommend a management decision and define the evidence threshold that would trigger it.

Expected answer / marking basis: Decision + measurable trigger.

LEVEL MODERATION & REPORTING

Recommended reporting: maintain the four domain scores separately as well as a weighted total. A student may have a strong overall total but still need targeted support in practical technique, data interpretation or problem solving.

Suggested achievement bands: 85-100 Advanced; 70-84 Secure; 55-69 Developing; 40-54 Emerging; below 40 Requires targeted support. Schools may adjust bands to their approved assessment policy.

Teacher moderation notes: